

Aleksandar Filipovic

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EDUCATION

University of British Columbia

Bachelor of Applied Science in Electrical Engineering

Vancouver, BC

Expected April 2029

- **CGPA: 83%** | Dean's Honour List
- **Relevant Coursework:** Circuit Analysis (**100%**), Integral Calculus (93%), Mechanics (92%), Electrostatics (89%)

EXPERIENCE

Software Engineering Intern

Rivian and Volkswagen Group Technologies

May 2026 – Dec. 2026

Vancouver, BC

- Engineered Software-in-the-Loop (SIL) frameworks in **Pytest** to simulate DIN and ISO charging sessions with parsed PCAP logs to spoof **100+** real-world payloads, accelerating bug identification and firmware validation.
- Developed automated load-strain tests for charging ECUs by spoofing comprehensive signal libraries with varying cycle times over Ethernet, accurately determining processing bottlenecks and **5+** hardware thresholds.
- Architected a VLAN tagging interface using Linux kernel sockets and routing tables, implementing TCAM switch rules to prioritize critical charging signals over heavy network traffic, reducing latency by **20%**.
- Resolved **20+** bug tickets by diagnosing vehicle communication issues using **CANalyzer** and standard CAN tools, implementing and validating robust firmware fixes within an embedded environment.

Embedded Designer

UBC BAJA SAE

Sept. 2025 – Present

Vancouver, BC

- Design firmware in **C/C++** for an off-road racing vehicle, managing I/O for **5+ sensors and actuators** on **STM32 micro-controllers** with minimal processing contention through optimized interrupts and DMA.
- Developed a continuous variable transmission (CVT) thermal monitoring system with real-time fault detection, defining process control parameters and shutdown thresholds to **prevent mechanical failure by 95%**.
- Created a signal conditioning circuit to convert inductive spark plug pulses into clean digital tachometer signals, enabling ECU-based RPM measurement up to **4500 RPM** and real-time transmission to the driver dashboard.
- Optimize **CAN bus** communication protocols across **6 distributed ECU** nodes for the vehicle's telemetry interface, ensuring reliable data transmission, robust fault handling, and sub-millisecond latency.

Electrical Engineering Intern

MCW Consultants LTD

May 2025 – July 2025

Vancouver, BC

- Generated **10+ detailed technical specifications** and drawing sets in AutoCAD and Revit, adhering strictly to Canadian Electrical Code standards to facilitate safe, scalable, and highly efficient power distribution frameworks.
- Analyzed gaps between as-constructed conditions and engineering designs for YVR Airport electrical systems, identifying **5+ discrepancies** and updating system documentation to reflect field conditions.
- Collaborated with senior engineers to update process standards, maintain change logs, and produce operations manuals and hand-over documents across **20+ electrical systems**.

PROJECTS

CVT Monitoring System | *STM32, Embedded C, Real-Time Control, CAN*

Sept. 2025 – Feb. 2026

- Wrote embedded C firmware on an **STM32 microcontroller** to read CVT belt temperature via an **IR sensor** through **I2C**, transmitting real-time data over CAN bus to a driver dashboard for pit-stop decision-making during endurance races.
- Designed sensor mounting hardware in SolidWorks, iterating through 3D-printed prototypes before fabricating the final mount in sheet metal for durability under race conditions.

FPGA Tron Game | *C, Nios V, State Machines, Real-Time Systems*

Nov. 2025

- Implemented complex **state-machine logic** on a DE10-Lite FPGA using C and a Nios V soft processor to manage real-time game states and control flow.
- Managed **memory-mapped I/O** to interface with peripherals; debugged interrupt handling and timing-critical control paths for precise system response.

TECHNICAL SKILLS

Embedded: C/C++, Embedded C, State Machines, Sensor Integration, Micro-controllers, data acquisition (DAQ)

Protocols: CAN Bus, Memory-Mapped I/O, UART, Serial Communication

Software: Python, PostgreSQL, Git, Solderworks, CAD, MATLAB, Altium Designer, Docker, Pytest

Hardware: Oscilloscope, Multimeter, Logic Analyzer, Digital & Analog Circuits, HIL/SIL Concepts